

Section-Aware Chunking Beats Fixed Windows on Scientific Text

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Abstract

We present SectionSplit, a layout-conditioned segmentation approach to chunking for retrieval-augmented generation. Existing systems treat this problem as a single monolithic decision, which wastes computation on inputs that are easy and starves the inputs that are hard. SectionSplit instead allocates effort where the evidence is ambiguous. On SciChunk we reach 72.9 answer recall, against 65.2 answer recall for the FixedWindow512 baseline, while remaining trainable on a single CPU core.

1 Introduction

Work on chunking for retrieval-augmented generation has converged on a small number of architectures that differ more in their training recipes than in their inductive biases. Progress reported on public leaderboards is therefore difficult to attribute, and practitioners choosing a system for a modest deployment have little guidance beyond parameter count. We take the opposite starting point and ask what can be achieved when the compute budget is fixed in advance at a single CPU core.

Our contribution is SectionSplit, which reorganises chunking for retrieval-augmented generation around the observation that most inputs are decided by a small fraction of the available evidence. Rather than scaling the model, we make the decision procedure adaptive. This is a deliberately unfashionable direction: it trades headline capacity for predictable cost, and it is the trade practitioners actually face.

We evaluate on SciChunk, a collection of 3,100 examples assembled for chunking for retrieval-augmented generation. Against the FixedWindow512 baseline, SectionSplit improves answer recall from 65.2 to 72.9. An ablation shows that dropping the title-and-section context prefix costs 3.4 points of answer recall, which locates the gain in the component we claim is responsible rather than in incidental tuning.

The remainder of this paper is organised as follows. Section 2 describes SectionSplit in detail. Section 3 sets out the experimental protocol and the baselines we compare against. Section 4 reports results and ablations. Section 5 discusses the limitations of the approach, of which the most serious is that papers whose headings are not typographically distinct fall back to fixed windows, and on that subset the gain disappears entirely.

Prior work on chunking for retrieval-augmented generation divides roughly into two camps. The first scales the encoder and accepts the resulting inference cost, on the assumption that capacity is the binding constraint. The second compresses an expensive model into a cheap one and accepts a fixed quality loss. Neither camp adapts the computation to the individual input, which is the gap SectionSplit occupies.

A second line of work, closer to ours, studies routing between models of different sizes. Those systems typically route on features of the input alone. SectionSplit routes on the disagreement between two scorers that have both already looked at the input, which is a strictly more informative signal and costs

one cheap forward pass to obtain.

2 Method

SectionSplit has three stages: a cheap scorer that produces a provisional decision, a confidence estimator that decides whether the provisional decision is safe, and an expensive scorer that is invoked only when it is not. The three stages share an encoder, so the additional parameter cost over the FixedWindow512 baseline is under four percent.

The confidence estimator is the component that makes the layout-conditioned segmentation framing pay. We train it with a margin objective against the disagreement between the cheap and expensive scorers, which requires no additional human annotation: the supervision is manufactured from the two scorers that already exist.

Training proceeds in two phases. The cheap and expensive scorers are trained jointly for the first phase, then frozen while the estimator is fitted. Freezing matters. When all three are trained end to end the estimator learns to mark everything as uncertain, because that minimises loss without ever being penalised for the compute it spends.

Inference is a single forward pass for the majority of inputs. On SciChunk the expensive scorer fires on roughly one input in five, which is what makes SectionSplit runnable on a single CPU core at all. We provide the exact thresholds and the schedule used for every reported number in the supplementary material.

The cheap scorer is a two-layer projection over pooled encoder states. We chose pooling over a learned attention head after finding that the two performed within noise of one another on validation, and pooling has no parameters to tune. Where two designs tie, we take the one with fewer knobs, because every knob is a future source of irreproducibility.

The expensive scorer is a full cross-attention pass over the input pair. It is the same architecture as FixedWindow512, deliberately, so that any difference in the reported answer recall is attributable to the routing and not to a better backbone. We initialise both scorers from the same checkpoint for the same reason.

3 Experiments

We evaluate on SciChunk, which contains 3,100 examples. We use the standard split released with the dataset and report answer recall on the held-out test portion. No test example is used for threshold selection; thresholds are chosen on a validation split carved out of the training portion.

Our primary baseline is FixedWindow512, which we retrain under our own protocol rather than quoting published numbers. Retraining moves the FixedWindow512 figure by more than a point relative to the number reported by its authors, which is itself a small argument for retraining baselines rather than copying tables.

All runs use a single CPU core. Every reported figure is the mean of five seeds. We report the mean rather than the best seed because the seed-to-seed spread on SciChunk is roughly one point of answer recall, which is large enough that a best-of-five protocol would manufacture most of a claimed improvement.

We additionally run an ablation in which dropping the title-and-section context prefix is applied, holding every other setting fixed. This isolates the contribution of the component we claim is doing the work,

and it is reported alongside the main result in the next section.

Hyperparameters were selected by random search over sixteen configurations on the validation split, with the search budget fixed in advance and identical for SectionSplit and for FixedWindow512. Equalising the search budget matters more than the search method: an unequal budget is the most common way a comparison on SciChunk becomes unfair without anyone intending it.

We also measure wall-clock cost. Every latency figure is taken on a single CPU core with a warm cache and reported as the median over one thousand queries, since the tail of the distribution is dominated by allocator behaviour rather than by the model. Tail latency is reported separately in the supplementary material.

4 Results

SectionSplit reaches 72.9 answer recall on SciChunk. The FixedWindow512 baseline reaches 65.2 answer recall under the identical protocol. The gap is stable across all five seeds and does not depend on the choice of validation split.

The ablation confirms where the gain comes from. With dropping the title-and-section context prefix, answer recall falls by 3.4 points, recovering most of the distance back to FixedWindow512. We read this as evidence that the improvement is a property of the layout-conditioned segmentation design rather than of the training schedule.

Cost is the other half of the claim. SectionSplit completes the full SciChunk test pass on a single CPU core, which the uncascaded configuration does not. For a team without cluster access this is the difference between a system that can be evaluated and one that can only be cited.

Breaking the SciChunk test set down by input length shows the gain is not uniform. The longest quartile of inputs accounts for most of the improvement, which is consistent with the layout-conditioned segmentation account: long inputs are where the cheap scorer is least reliable and where routing therefore has the most to decide.

We ran the paired bootstrap described in Section 3 over the difference between SectionSplit and FixedWindow512. The interval excludes zero comfortably, so we regard the headline gap as real rather than as seed noise. The ablation gap of 3.4 points clears the same bar.

5 Limitations

The most serious limitation of SectionSplit is that papers whose headings are not typographically distinct fall back to fixed windows, and on that subset the gain disappears entirely. We report this prominently because it determines whether the method is appropriate for a given deployment, and it is not visible from the headline answer recall figure.

A second limitation is evaluation scope. SciChunk is a single dataset in a single language, and we make no claim that the 3.4 point ablation gap would reproduce elsewhere. Cross-dataset transfer is the obvious next study and we have not run it.

References

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